

Birla Institute of Technology and Science, Pilani
Quantum Machine Learning
(AIMLCZG545)

**Assignment 1: Quantum Circuit Design and
Problem Exploration**

Submission Deadline: 14 June 23:59 PM

Instructions

- Total Marks: **10**
- Submit:
 - Qiskit notebook (.ipynb) with code AND answers to Part A questions. (7 Marks)
 - Screenshot of performing the assignment on BITS Virtual Lab. (0.5 Marks)
 - PDF report for Part B. (2.5 Marks)
- Clearly show circuits, outputs, and reasoning
- **All questions in Part A are compulsory**

Part A: Quantum Circuit Design (7.5 Marks)

Objective: Analyze and design quantum circuits to achieve specific behaviors.

This assignment has to be performed on BITS Virtual Lab and ONE screenshot of the same has to be attached (0.5 Marks)

Problem 1: Controlled Probability Design (2 Marks)

Design a single-qubit circuit such that:

$$P(0) \approx 0.75, \quad P(1) \approx 0.25$$

Constraints:

- Initial state: $|0\rangle$
- Allowed gates: H, X, R_y, R_z

Questions:

- a) What rotation angle did you choose? Justify mathematically.
- b) Draw the quantum circuit.
- c) Show the measurement results.
- d) Explain how the circuit produces the desired probabilities.

Problem 2: Phase vs Measurement (1.5 Marks)

Design two circuits such that:

- The final states differ due to phase
- Measurement probabilities are identical

Questions:

- a) Draw both circuits.
- b) Show output distributions.
- c) Are the states physically identical? Explain.
- d) Why does measurement fail to distinguish between them?

Problem 3: Entanglement Analysis (1.5 Marks)

Consider:

Circuit 1: Apply H followed by CNOT

Circuit 2: Apply only H

Questions:

- a) Implement both circuits.
- b) Compare measurement outputs.
- c) Which circuit produces entanglement? Justify.
- d) Explain the role of CNOT in creating correlation.

Problem 4: Reverse Engineering (2 Marks)

Given:

$$P(00) \approx 0.5, \quad P(11) \approx 0.5$$

Questions:

- a) Design a circuit that produces this distribution.
- b) Draw the circuit.
- c) Verify using simulation.
- d) Explain why your circuit produces this result.

Part B: Literature Survey (3 Marks)

Objective: Identify a classical problem with scaling limitations.

Questions:

- a) What is the problem? Describe briefly.
- b) Why is it difficult to solve classically?
- c) What are the key limitations (e.g., time, data, dimensionality)?
- d) Provide at least 2 references.
- e) Use the given Latex template to prepare the report.

Note: Emphasis will be on clarity of reasoning, correctness of circuits, and conceptual understanding.

For any other queries / clarifications related to the assignment, please write on [Taxila Discussion Forum, after checking existing queries and their answers.](#)

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This is a Hard Deadline - No extensions will be provided.

All the Best !!
